SPATIAL AI + COMPUTING · COLUMBIA GSAPP

Positioning + Movement

A food-delivery semantic model of Morningside Heights, with live prepositions of place and movement

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Simulation: ishika100603.github.io/Spatial_ai/food-delivery-3d.html



The simulation: one city block grid, many occupants

SITE

W 111th–116th St

Broadway to Morningside Dr, ≈ 420 m × 405 m, real street names

AGENTS

3 riders

Motorbike, car, e-bike; 4 restaurants, 3 customers, food parcels

TRAFFIC

82 vehicles

160 pedestrians, 21 traffic lights; numbers set live with sliders

TIME

Live clock

Orders, cooking, signals and rush hour run on one simulated clock

Each position and movement on the next slides is recognised live and marked in the 3D scene.

What the model recognises

POSITIONING

IN / ON · the parcel IN the bag, ON the pickup shelf

ADJACENT · the rider right beside the door

BETWEEN · someone between the rider and its turn

AMONG · a rider among a crowd

BEHIND · a rider following a car in its lane

MOVEMENT

TOWARD / AWAY FROM · the customer’s door

INTO / OUT OF · the bag, a crowd

PAST · restaurants, doors, other riders

AROUND · a closed street

Each one draws its own marker in the 3D scene as it happens.

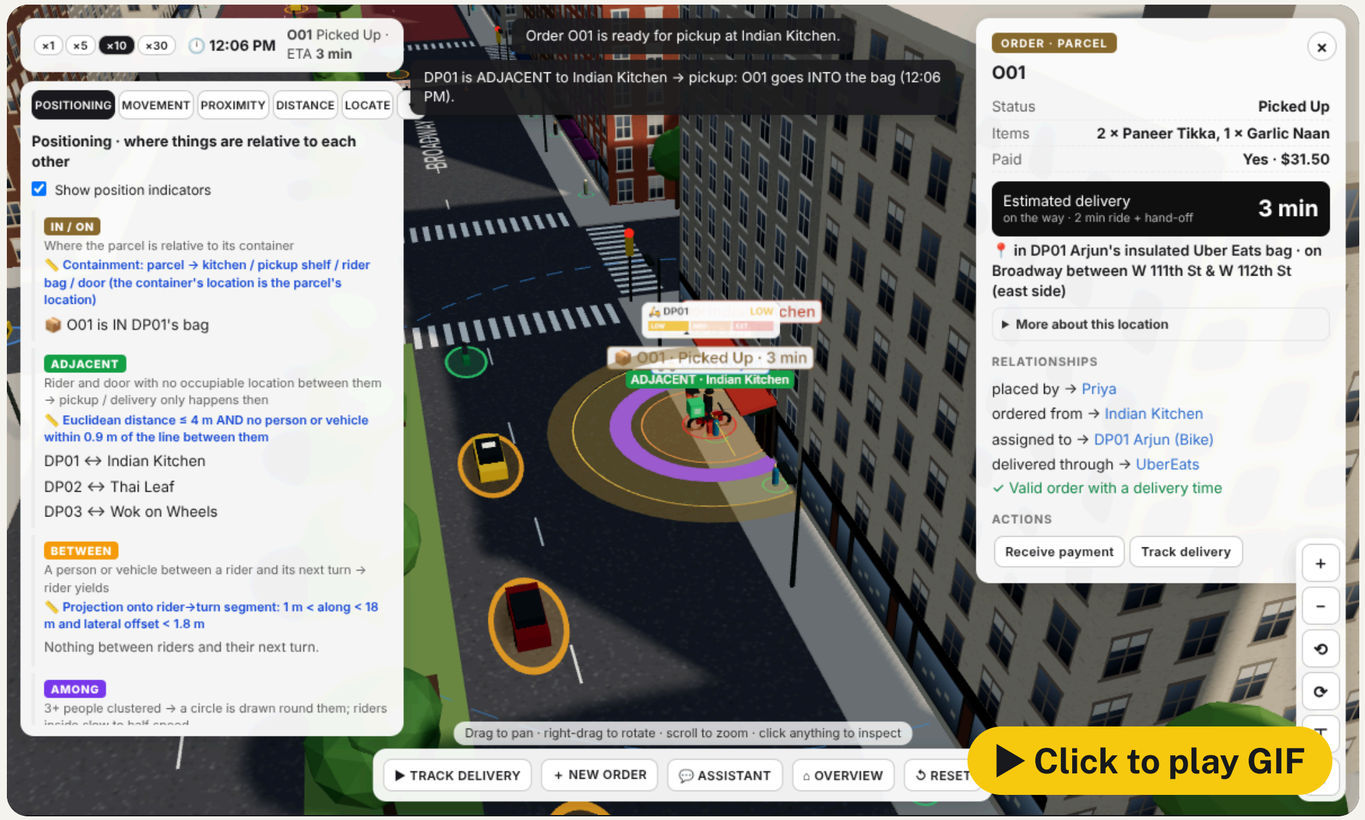
Picking up the parcel

ADJACENT · DP01 pulls up right beside Indian Kitchen’s door

INTO · the restaurant hands O01 over, into DP01’s bag

NOT ADJACENT · if someone stands between them, the hand-over waits

A green link joins rider and door while they are adjacent.



A crowd at the crossing

AMONG · people waiting for the walk signal bunch together and get a purple circle

INTO / OUT OF · a rider entering the circle slows to half speed until it leaves

Rush hour at Broadway & W 114th St.



Behaviours that fire when a placement is met

A · PLACEMENT → SYSTEM RESPONDS

AMONG: 3+ people close together → a circle is drawn round them, and riders inside it slow to half speed

BETWEEN: someone in the rider's corridor → the rider yields

BEHIND: vehicle ahead in the lane → the rider keeps a following gap

B · INTRINSIC CAPABILITIES

Rider · "deliver": the hand-over happens only when ADJACENT to the customer's door

Restaurant · "hand over": releases the parcel INTO an ADJACENT rider's bag

Barricade · "warn": warns riders nearby, who re-route AROUND it

One delivery, logged as movement

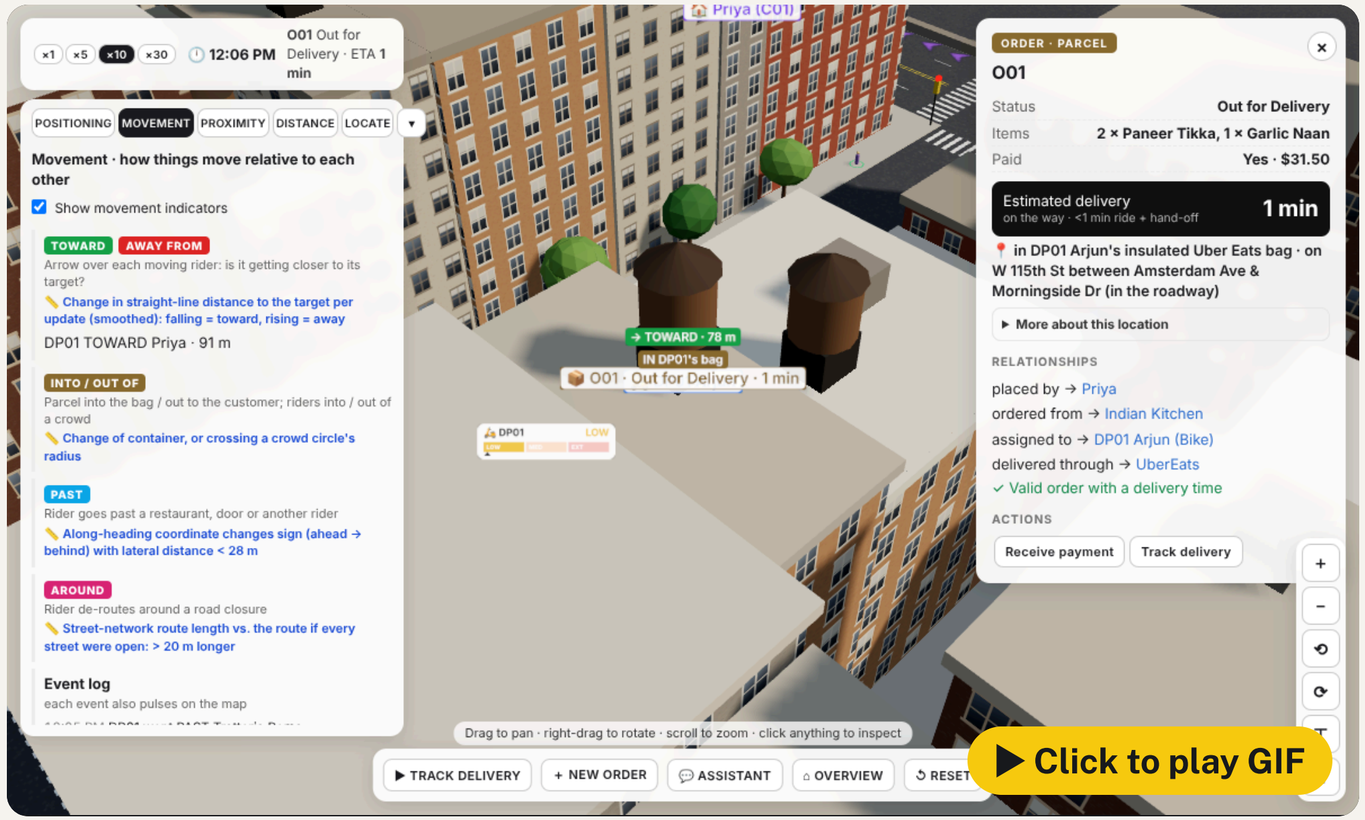
INTO · O01 goes into DP01's bag, out of Indian Kitchen

TOWARD · green arrow while DP01 heads for Priya's door

PAST · DP01 goes past Sam's door and Trattoria Roma

OUT OF · O01 leaves the bag, into Priya's hands, once ADJACENT

Each event pulses on the map and is time-stamped in the MOVEMENT tab.

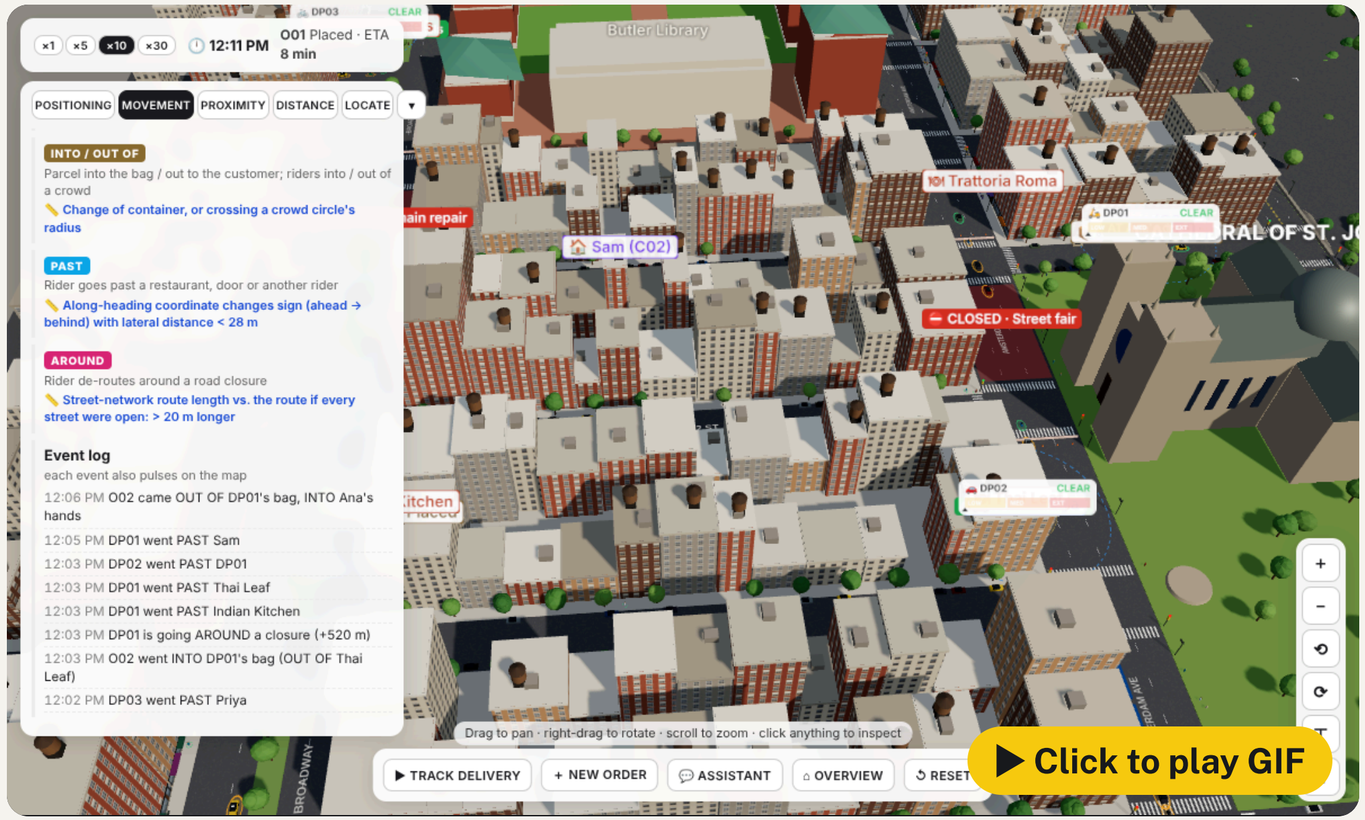


Going around a street fair

WARN · Amsterdam Ave is closed at W 112th–113th St; the barricade warns riders nearby

AROUND · Thai Leaf's rider takes Broadway instead to reach Ana, a 520 m detour

The detour is labelled over the closure: "↻ going AROUND".



Try it

ishika100603.github.io/Spatial_ai/food-delivery-3d.html

1. Open the **POSITIONING** tab: every notion with its distance metric, live.
2. Press ▶ **TRACK DELIVERY**, then open **MOVEMENT** to watch the events.
3. Drag to pan, right-drag to rotate, scroll to zoom; ⌂ returns to the whole site.

Justin & Ishika · Spatial AI + Computing · Columbia GSAPP · code: github.com/ishika100603/Spatial_ai